

$$1) (-3)^2 - [(-30) : 3 + 2 \cdot (-5) - 8 \cdot (2-5)] \cdot 2 - 9 = ?$$

$$9 - [-10 - 10 - 8 \cdot (-3)] \cdot 2 - 9$$

$$9 - [-20 + 24] \cdot 2 - 9$$

$$9 - 4 \cdot 2 - 9$$

$$9 - 8 - 9 = -8$$

$$2) x - (1 - [2x - 3(x+1)]) + 4(2-2x) = ?$$

$$x - (1 - (2x - 3x - 3)) + 8 - 8x$$

$$x - (1 - (-x - 3)) + 8 - 8x$$

$$x - (1 + x + 3) + 8 - 8x$$

$$x - (x + 4) + 8 - 8x$$

$$x - x - 4 + 8 - 8x = 8x + 4$$

$$3) -3[(x+2y) - 3 - 2(x+y)] - [(3x+1) - (x-8)] = ?$$

$$-3(x+2y-3-2x-2y) - (3x+1-x+8)$$

$$-3(-x-3) - (2x+9)$$

$$3x+9-2x-9 = x$$

$$4) 2\frac{3}{4} \cdot (-3\frac{5}{6}) \cdot \frac{24}{11} = ?$$

$$(1\frac{1}{4}) \cdot (-\frac{23}{2}) \cdot (\frac{24}{11}) = -23$$

$$5) (2\frac{1}{3} + 1\frac{1}{4}) - (1\frac{1}{2} - 1\frac{1}{4}) = ?$$

$$(\frac{7}{3} + \frac{5}{4}) - (\frac{3}{2} - \frac{5}{4})$$

$$(\frac{28}{12} + \frac{15}{12}) - (\frac{6}{4} - \frac{5}{4})$$

$$\frac{28+15}{12} - \frac{(6-5)}{4} = \frac{43}{12} - \frac{1}{4} = \frac{40}{12} = \frac{10}{3}$$

$$6) \left[\left(-\frac{3}{4} \right) : \left(\frac{2}{3} \right) \right] \cdot \left[\left(\frac{5}{3} \right) \cdot \left(-\frac{4}{3} \right) \right] = ?$$

$$\left(-\frac{3}{4} \cdot \frac{3}{2} \right) \cdot \left(-\frac{20}{9} \right) =$$

$$\left(-\frac{9}{8} \right) \cdot \left(-\frac{20}{9} \right) = \frac{20}{8} = \frac{10}{4} = \frac{5}{2}$$

$$7) \left(\frac{2}{3} \right)^{-1} + \left(\frac{2}{7} \right)^{-1} = ?$$

$$\left(\frac{3}{2} \right)^{-1} + \frac{7}{2} = \frac{3}{5} + \frac{7}{2} = \frac{6+35}{10} = \frac{41}{10}$$

(2) (5)